ModelIt!

Student Activity Pack

How Do Seasons Change?

The Science of Seasonal Patterns

Grade 1 | NGSS 1-ESS1-2 | Patterns

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Draw your favorite season. Show what the weather is like and what you wear outside.

2 Circle the season when the sun shines the longest each day: spring, summer, fall, winter.

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Amount of Daylight

Temperature

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Summer Days

Amount of Daylight set to very high (long sunny days)

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Winter Days

Amount of Daylight set to very low (short sunny days)

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Why is it hot in summer and cold in winter? Use evidence from your simulation to support your answers.

8 STEM Challenge: Create a Season Wheel

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YOUR ENGINEERING MISSION

Build a spinning season wheel that shows what happens to daylight and temperature in each of the four seasons.

THE SCENARIO

A weather station needs a tool to help explain seasons to kids. Design a season wheel that shows how daylight and temperature change from spring to summer to fall to winter. Make it spin to show the pattern!

Engineering Design Thinking

1. HOW WILL YOU DIVIDE YOUR WHEEL INTO FOUR EQUAL PARTS FOR FOUR SEASONS?
2. WHAT COLORS WILL YOU USE TO SHOW WARM AND COLD TEMPERATURES?
3. HOW WILL YOU SHOW THAT SUMMER HAS MORE DAYLIGHT THAN WINTER?
4. HOW WILL YOU MAKE YOUR WHEEL SPIN TO SHOW THE PATTERN REPEATING?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Meteorologists study weather patterns and seasons to help people prepare for storms, heat waves, and cold snaps. They earn $50,000-$100,000/year.